

GNN-Based BERT for Understanding Context from Music

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Abstract—This project studies music-context prediction using text and audio structure. DistilBERT is used to encode captions or tag-based context, while GraphSAGE encodes graphs whose nodes are short audio segments. Three required experiments are evaluated: a text-only MusicCaps tag classifier, an audio-only GraphSAGE model compared with a spectrogram CNN on GTZAN, and a four-way GNN-BERT ablation on MagnaTagATune. A fourth experiment trains a contrastive audio-caption model on recovered real MusicCaps intervals and evaluates bidirectional retrieval. The implementation uses frozen data splits, validation-only model selection, and saved evaluation artifacts. The main goal is not to claim state-of-the-art performance, but to test whether graph and text representations provide complementary information under a reproducible experimental setup.

I. INTRODUCTION

Music context includes genre, mood, instrumentation, rhythm, and the words people use to describe a recording. A spectrogram CNN can learn local time-frequency patterns, but it does not directly model relationships between distant parts of a track. A graph provides a simple way to represent those relationships: each node is an audio segment and edges represent either temporal adjacency or acoustic similarity. Text encoders provide a second view of the same recording through captions or tags.

The project follows the supplied CSE425 assignment. Task 1 tests a BERT baseline, Task 2 tests an audio graph, and Task 3 combines both modalities. The main research question is whether the fusion models improve over their single-modality ablations. Task 4 is treated as an advanced extension because MusicCaps audio must be recovered from external YouTube links and some clips may no longer be available.

A. Relation to the CSE425 Course Outline

The work also follows the supplied CSE425 Neural Networks course outline. It applies CNNs to log-mel spectrograms, transformer attention through DistilBERT, transfer learning through pretrained text weights, and AdamW optimization with gradient clipping, dropout, learning-rate scheduling, and early stopping. Ablations, confusion matrices, per-tag scores, and attention visualizations support model interpretation and transparent comparison. The GNN is a project-specific architecture required by the separate project brief. Together, the stages form an end-to-end workflow covering data preparation, training, validation, evaluation, and explainability.

II. DATASETS AND SPLITS

A. MusicCaps

MusicCaps contains 5,521 expert-written music captions. Rows marked as AudioSet evaluation examples are held out as the test set. The remaining rows are split deterministically into 90% training and 10% validation data. This gives 2396 training, 267 validation, and 2858 test examples. The Task 1 label vocabulary is built from the training split only, with a minimum frequency of 20.

B. GTZAN

GTZAN contains ten genres with approximately 100 tracks per genre. One unreadable track is excluded during preprocessing and recorded in the preprocessing report. Because GTZAN contains repeated files, SHA-256 hashes are used as group identifiers during a seeded, per-genre split. All copies of identical audio remain in one partition. The split audit found 14 duplicate groups and zero groups crossing train, validation, and test. Both Task 2 models use exactly the same IDs.

C. MagnaTagATune

MagnaTagATune contains 29-second audio clips with binary tags. The official archive-shard protocol is used: shards 0–b are training, shard c is validation, and shards d–f are test. After checking usable graph files, Task 3 contains 18706 training, 1825 validation, and 5329 test clips. Exact clip overlap and exact (artist, title, album) overlap are zero. Some artist overlap remains and is reported in the split audit rather than being hidden.

III. PREPROCESSING

All audio is loaded as mono and resampled to 22,050 Hz. For the CNN, a 128-bin log-mel spectrogram is computed and normalized per track. Graph nodes use non-overlapping five-second segments. Each node has 77 summary features: MFCC means and standard deviations, chroma means and standard deviations, spectral contrast, spectral centroid, zero-crossing rate, and RMS energy. Features are standardized within each track.

For node features x_i and x_j , an acoustic edge is added when

$$\frac{x_i^\top x_j}{\|x_i\| \|x_j\|} > \tau, \quad (1)$$

where $\tau = 0.5$. Consecutive segments are always connected in both directions, and self-loops are added. Every saved graph is checked for non-finite features, empty node sets, and invalid edge indices. A graph coherence score $S = \frac{1}{|E'|} \sum_{(i,j) \in E'} \mathbf{1}[\cos(x_i, x_j) > \tau]$ is also computed over non-self-loop edges. Across GTZAN and MTAT, coherence averages 0.012 and 0.030 respectively, confirming that most edges are temporal-adjacency links between acoustically distinct segments rather than similarity-based shortcuts.

Text is tokenized with the uncased DistilBERT tokenizer. Task 1 uses the expert caption. Task 3 constructs a short sentence from 31 instrument/vocal context tags. The 19 prediction targets contain 12 genre and 7 mood tags. The two tag sets are lexically disjoint. They still come from the same annotation process, so indirect statistical relationships are possible.

IV. MODELS

A. Task 1: BERT Baseline

The DistilBERT representation of the first token is passed through a two-layer classifier. For target vector y and logits s , training uses binary cross-entropy with logits. Class weights are calculated from training labels only. The classifier head is trained first with the BERT backbone frozen, followed by full-backbone fine-tuning.

B. Task 2: GraphSAGE and CNN

The GNN uses three GraphSAGE layers with 256 hidden units, followed by global mean pooling and a classification head. For graph G with final node states h_i , the graph representation is

$$g = \frac{1}{|V|} \sum_{i \in V} h_i. \quad (2)$$

The comparison model is a three-block convolutional network over the normalized log-mel spectrogram. Both models use cross-entropy loss and validation Macro-F1 for checkpoint selection.

C. Task 3: GNN-BERT Fusion

Four conditions are trained: BERT only, GNN only, early concatenation, and cross-attention. For cross-attention, the graph embedding is the query and DistilBERT token embeddings are keys and values:

$$Q = gW_Q, \quad K = H_{text}W_K, \quad V = H_{text}W_V, \quad (3)$$

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad (4)$$

$$z = [g; AV]. \quad (5)$$

The fused vector z is passed to a multi-label classifier. Padding positions are masked before the softmax. BERT is first frozen and then its last transformer layer is fine-tuned.

D. Task 4: Contrastive Extension

The extension projects GraphSAGE and caption encodings into a shared 128-dimensional normalized space and uses a symmetric InfoNCE loss. DistilBERT is frozen, while GraphSAGE is initialized from the Task 3 audio encoder. It is evaluated with caption-to-audio and audio-to-caption Recall@1, Recall@5, and Recall@10.

V. EXPERIMENTAL SETUP

Experiments use seed 42. AdamW, gradient clipping, and early stopping are used for all neural models. Classification checkpoints are selected with validation Macro-F1. Task 4 instead uses mean bidirectional validation Recall@5. A global multi-label threshold is then selected on validation predictions and applied once to test predictions. The report uses Macro-F1, Micro-F1, and mean per-tag AUC-PR. GTZAN additionally reports accuracy and a confusion matrix.

A. Training and Fine-Tuning Procedure

The same validation-first procedure is used for each model. First, the fixed split is loaded and class weights are calculated from training labels only. For every epoch, the model is updated on training batches, then evaluated on the validation partition. The checkpoint is replaced only when validation Macro-F1 improves, and early stopping ends a run after the stated patience. For a multi-label task, the best checkpoint produces validation probabilities used to choose one global decision threshold. Finally, that frozen checkpoint and threshold are evaluated once on the test partition. This order prevents test labels from influencing training, checkpoint selection, or threshold selection.

For Task 4, the requested MusicCaps ranges are frozen before the final run: records 0–399 of the training split, 0–99 of validation, and 200–299 of the official test split. Clips whose source video cannot be recovered are omitted without replacement. This leaves 364/89/93 usable real graph-caption pairs. The final test block was not used by the earlier feasibility run and is evaluated only after validation selects the checkpoint.

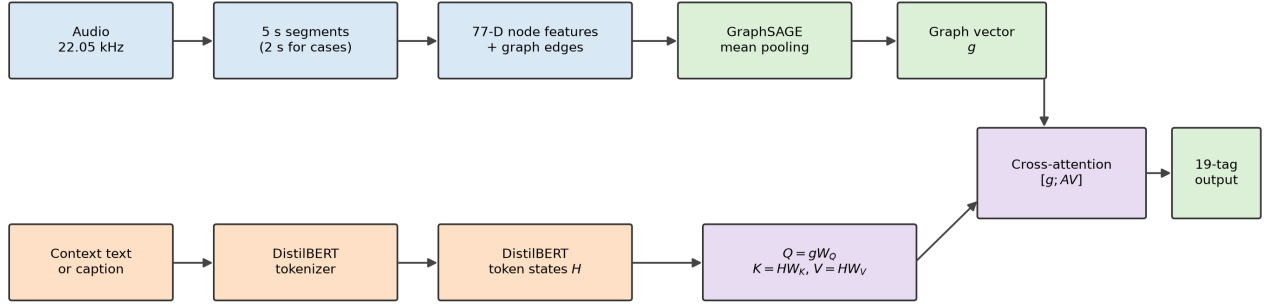
For DistilBERT fine-tuning, the transformer is initially frozen while the new prediction head and, for Task 3, the graph/fusion modules learn. The best frozen-stage checkpoint is reloaded before fine-tuning. Task 1 then updates the full backbone at learning rate 2×10^{-5} ; Task 3 updates only the final transformer layer at the same learning rate. Gradients are clipped to norm 1.0. This two-stage schedule was used because full transformer training from the first epoch was less stable on the available GPU.

VI. RESULTS

A. Task 1

The Task 1 test scores are Macro-F1 0.2863, Micro-F1 0.4000, and mean AUC-PR 0.2979. The difference between Macro-F1 and Micro-F1 is expected because the MusicCaps proxy vocabulary remains imbalanced even after the frequency filter.

Implemented GNN--BERT cross-attention pipeline



Ablations bypass one branch (BERT-only/GNN-only) or replace cross-attention with concatenation.

Fig. 1. Implemented audio-graph and text cross-attention pipeline.

TABLE I
MAIN EVALUATION TABLE. DASHES INDICATE METRICS THAT DO NOT APPLY.

Model	Accuracy	Macro-F1	AUC-PR	R@5
Random tags (MTAT)	—	0.1107	0.0662	—
Task 1: DistilBERT	—	0.2863	0.2979	—
Task 2: GraphSAGE	0.3495	0.3364	—	—
Task 2: CNN	0.7864	0.7859	—	—
Task 3: BERT only	—	0.2812	0.2525	—
Task 3: GNN only	—	0.2967	0.2442	—
Task 3: Concatenation	—	0.3708	0.3541	—
Task 3: Cross-attention	—	0.3786	0.3545	—
Task 4: Contrastive	—	—	—	0.0968

only 11.7% accuracy and 0.12 Macro-F1, confirming that both the GNN and CNN architectures learn meaningful representations beyond what a shallow classifier can extract from the same features. The confusion matrices show which genres account for the model difference. This comparison also tests a limitation of the graph representation: reducing a five-second segment to summary statistics discards more local detail than the full spectrogram.

C. Task 3

Early concatenation gives the highest held-out result, with Macro-F1 0.3708 and mean AUC-PR 0.3541. Cross-attention is second with Macro-F1 0.3786 and mean AUC-PR 0.3545. Both fusion models improve over the BERT-only and GNN-only conditions, which supports the claim that their information is complementary. However, cross-attention does not improve over simple concatenation. A likely reason is that the context sentences are short, templated lists of tags, so pooled features are already informative and the additional attention parameters do not provide a consistent benefit. Per-tag results are retained because an average score can hide weak performance on rare labels.

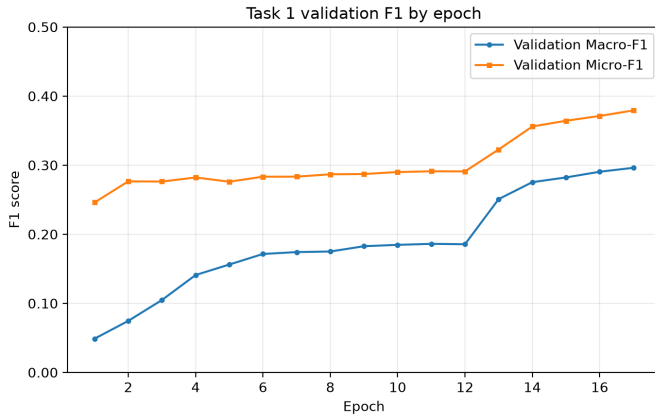


Fig. 2. Task 1 validation Macro-F1 and Micro-F1 across epochs.

B. Task 2

GraphSAGE obtains accuracy 0.3495 and Macro-F1 0.3364, while the CNN obtains accuracy 0.7864 and Macro-F1 0.7859. As an additional baseline, a PCA+MLP model is trained on aggregated hand-crafted node features (mean, standard deviation, min, max of the 77-D segment vectors). It achieves

TABLE II

FIRST THREE HELD-OUT TASK 1 PREDICTIONS IN FROZEN TEST ORDER. PREDICTED TAGS ARE THE FOUR HIGHEST-PROBABILITY OUTPUTS; TRUE TAGS ARE LIMITED TO THE TRAINING-DERIVED VOCABULARY.

Caption excerpt	True tags	Top predicted tags
The low quality recording features a ballad song that contains sustained strings, mellow piano melody and soft f... a male voice is singing a melody with changing tempos while snipping his fingers rhythmically. The recording sou... This song contains digital drums playing a simple groove along with two guitars. One strumming chords along with...	low quality, soulful, sustained strings melody, sad amateur recording, reverb medium tempo, piano, acoustic guitar, e-bass	emotional, passionate, low quality, mellow uptempo, medium tempo, amateur recording, medium to uptempo acoustic drums, e-bass, piano, medium to uptempo

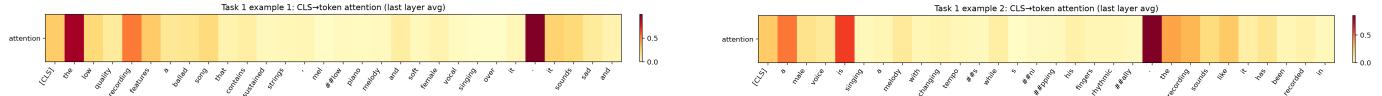


Fig. 3. CLS-to-token attention (last-layer head average) for two held-out Task 1 captions. Higher intensity indicates tokens receiving more attention from the classification token. The model attends most to genre and mood descriptors.

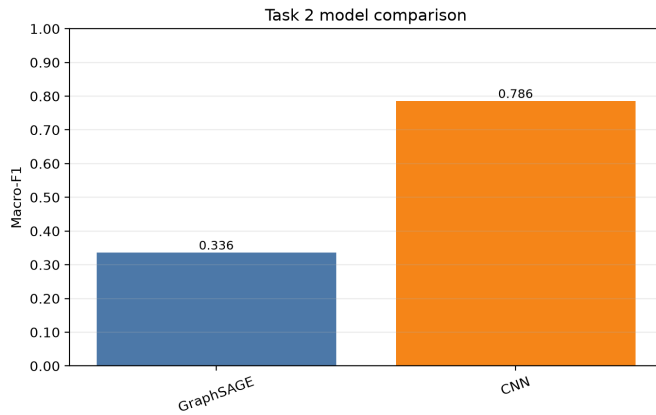


Fig. 4. Task 2 Macro-F1 comparison using the same GTZAN split.

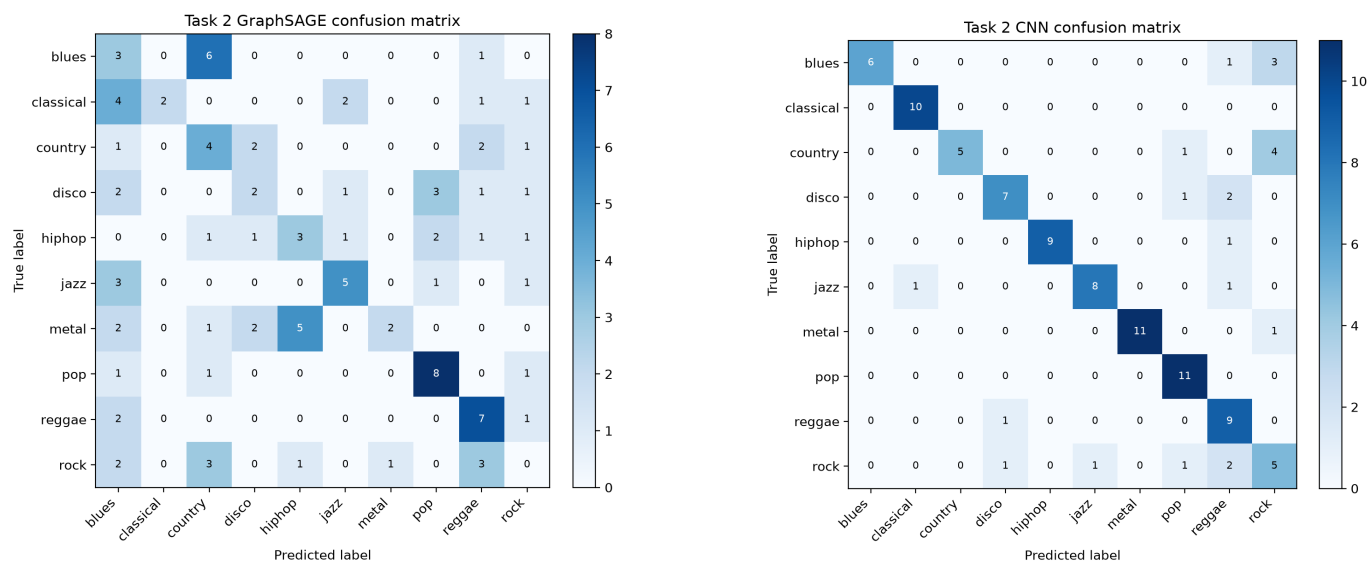


Fig. 5. Task 2 test confusion matrices for GraphSAGE (left) and CNN (right).

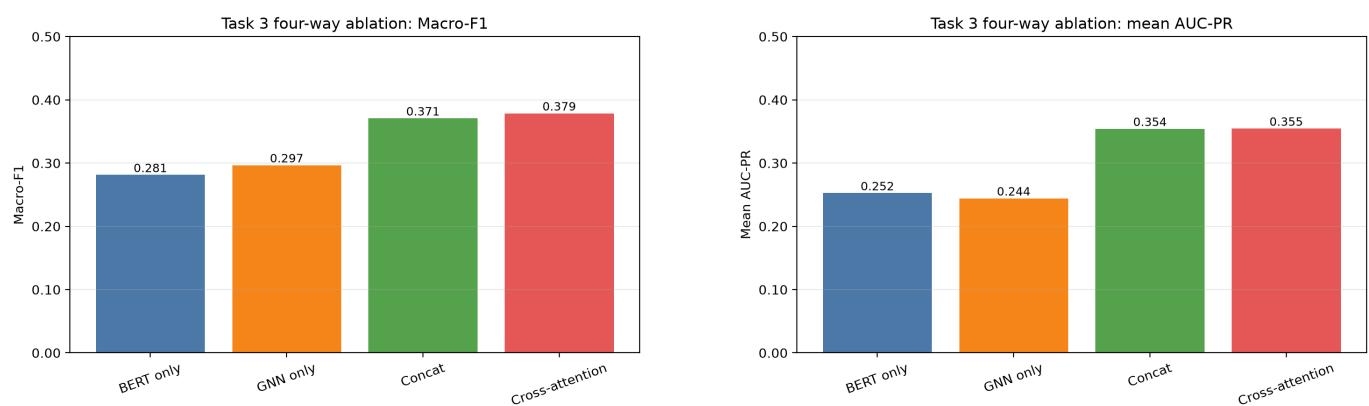


Fig. 6. Task 3 four-way ablation measured by Macro-F1 and mean AUC-PR.

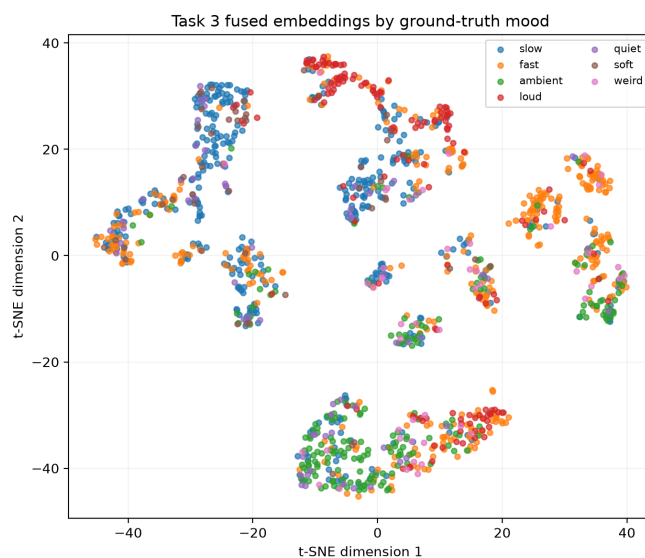
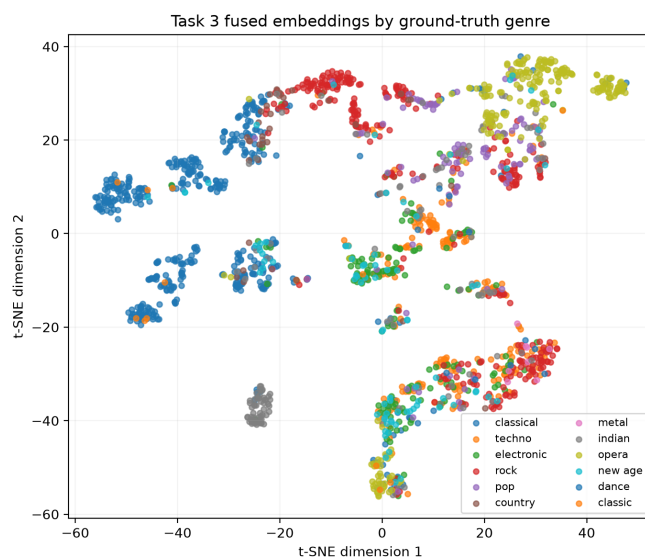
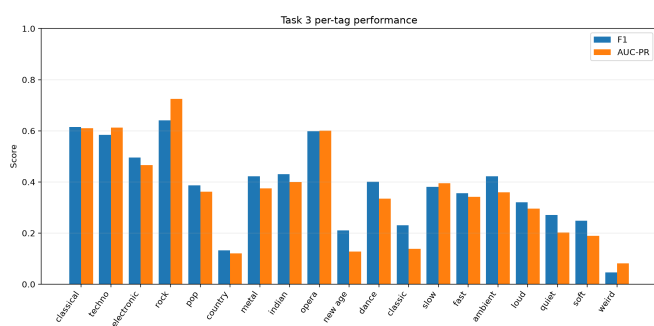


Fig. 7. Task 3 diagnostics: held-out per-tag scores (left) and ground-truth-labelled genre/mood t-SNE views (right).

D. Task 4

The selected contrastive checkpoint is from epoch 22. Caption-to-audio Recall@1/5/10 is 0.0215/0.0968/0.1720, and audio-to-caption Recall@1/5/10 is 0.0430/0.1398/0.2151. Random-ranking Recall@5 for this test size is 0.0538. The learning plot separates loss from retrieval recall and starts both vertical axes at zero, so changes remain readable without a misleading dual axis.

VII. QUALITATIVE ANALYSIS

Three supplementary cases use real held-out MusicCaps intervals and their expert captions. The MTAT-trained cross-attention checkpoint is applied without additional MusicCaps tuning. Each panel shows the temporal segment path, any similarity edges passing the fixed threshold, and the caption tokens receiving the most mean cross-attention. These visualizations are a qualitative transfer check, not a MusicCaps performance estimate or a causal explanation.

Task 4 additionally saves ten held-out captions with their top three retrieved audio clips. These examples preserve the clip IDs, source intervals, similarity scores, and whether the paired clip was retrieved. For the requested zero-shot comparison, caption and 19 tag-prompt embeddings are matched without Task 4 tag supervision. Its Macro-F1 is 0.0111, compared with 0.0500 when the supervised Task 3 cross-attention classifier is transferred directly to the same MusicCaps examples. Both are transfer checks because the available MusicCaps aspects are only an imperfect match to the MTAT target vocabulary.

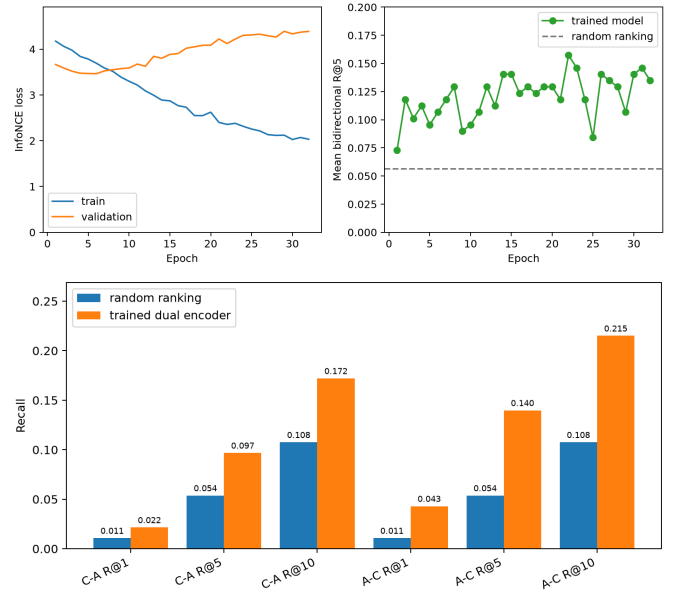


Fig. 8. Task 4 optimization (top) and held-out bidirectional retrieval compared with random ranking (bottom).

Task 3 qualitative transfer: real held-out MusicCaps graphs and expert captions

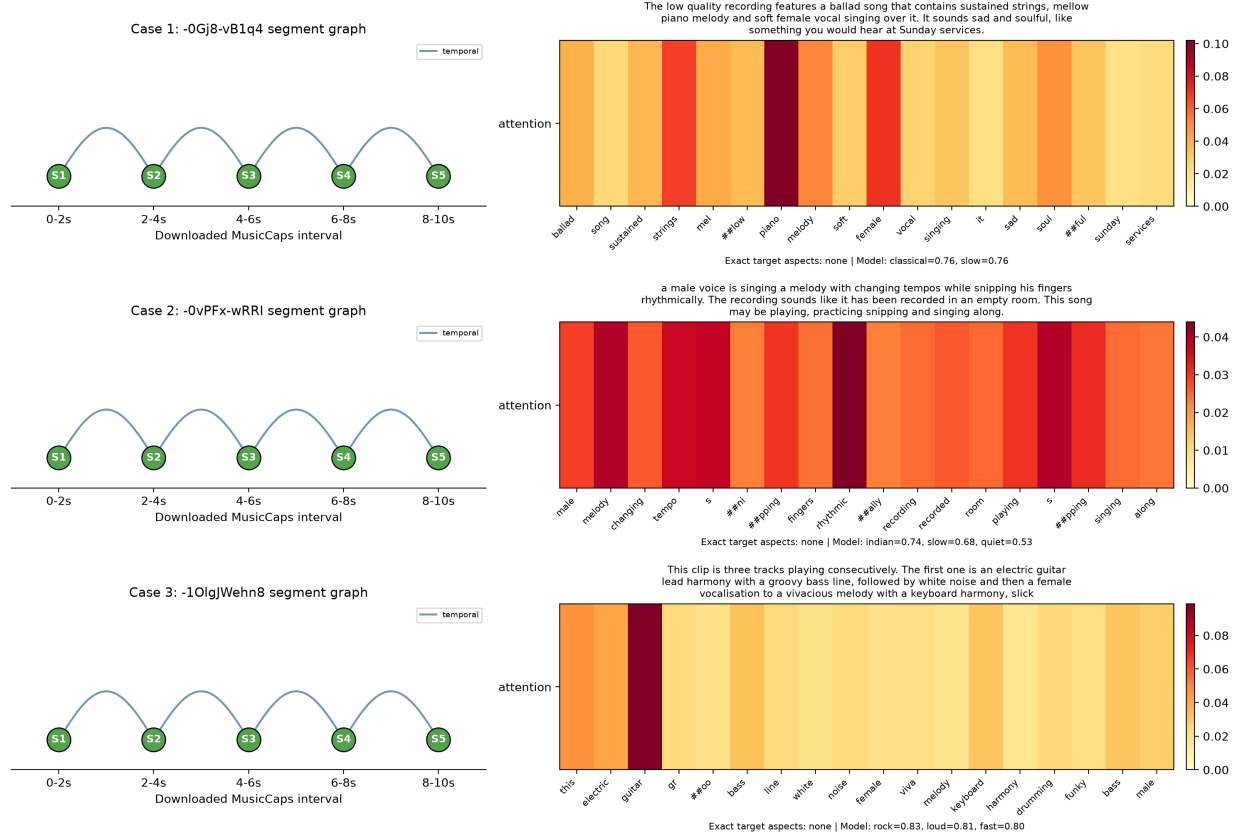


Fig. 9. Three deterministic held-out MusicCaps graph/caption alignment cases. Only the 18 highest-attention content tokens are displayed, in their original sequence order.

VIII. LIMITATIONS AND RESPONSIBLE USE

MagnaTagATune tags are noisy and were collected through a game. Context and target tags are disjoint in wording but are not independent. Artist overlap also remains in the official shard split. MusicCaps captions contain subjective descriptions, and YouTube availability can change the usable Task 4 sample. The contrastive experiment therefore represents a fixed recovered subset, not the complete MusicCaps benchmark. GTZAN is small and contains known data-quality concerns. For these reasons, the models should be treated as course experiments rather than general-purpose music understanding systems.

The project processes only public research datasets. Audio is not redistributed in the source repository. Generated predictions may reflect dataset biases and should not be used to make decisions about artists or listeners.

IX. CONCLUSION

This project implements text-only, graph-only, and multi-modal music-context models in a common PyTorch pipeline. On GTZAN, the CNN clearly outperforms GraphSAGE, showing that the 77-D segment summaries discard information

retained by the full mel spectrogram. On MTAT, both multi-modal models outperform their single-modality ablations, and early concatenation gives the strongest overall result. Cross-attention remains competitive but does not beat the simpler fusion method. The real-audio contrastive extension also supplies bidirectional retrieval, ten top-three examples, and a zero-shot tag-transfer check; its scores are interpreted against random ranking rather than presented as benchmark performance. Frozen splits, validation-only selection, duplicate audits, saved checkpoints, and generated tables make these findings reproducible without overstating them.

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